

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, C, C#

Machine Learning: Pandas, NumPy, Scikit-learn, EDA, Feature Engineering, Classification, Regression, Decision Trees, Predictive Modeling, Statistical Analysis, Anomaly Detection, Hyperparameter Tuning, Cross-Validation, Model Evaluation

Big Data: Databricks, Apache Spark, PySpark, Spark SQL, Delta Lake, Databricks SQL, Lakehouse Architecture, ETL/ELT, Data Pipelines, Data Migration, Batch Processing, Real-Time Processing, Workflow Automation, Apache Airflow

Streaming & Analytics: Apache Kafka, Apache Flink, Flink SQL, Stream Processing, Event-Time Processing, Windowing, Watermarking, Real-Time Analytics

Databases: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, ClickHouse, NoSQL, OLAP

Cloud & Tools: AWS, Amazon S3, Amazon EC2, Amazon RDS, Docker, Linux, Git, GitHub, Grafana, Streamlit

PROFESSIONAL EXPERIENCE

Data Science Intern, InsOps Inc., Chicago, Illinois

Jul 2024 – Present

- Contributed to InsOpsAI, a domain-specific SLM fine-tuned on insurance data, supporting use cases like Litigation Mitigation, analyzing historical claims and outcomes to predict case trajectories and inform proactive response strategies
- Performed exploratory data analysis and feature engineering on historical claims data to identify patterns correlating case attributes with litigation outcomes, feeding predictive modeling for Litigation Mitigation
- Built and validated data pipelines (Python, SQL, Databricks, Apache Airflow) to clean, transform, and prepare large-scale claims data from SQL Server, PostgreSQL, and AWS S3 into model-ready datasets for fine-tuning
- Implemented metadata-driven automation for schema discovery and transformation across heterogeneous sources, improving reliability of model training inputs
- Leveraged distributed processing and lakehouse architecture (Spark, Delta Lake) to scale feature computation and dataset preparation for the insurance-specific language model

Project Engineer, Soliton Technologies, India

Jul 2022 – Jun 2024

- Developed a C# WPF desktop application integrating C#, Python, and LabVIEW to streamline test automation workflows.
- Built reusable software components and automated workflows using object-oriented design, improving development and testing efficiency
- Developed customized solutions for Medtronic, automating ~94% of manual test cases while meeting medical-device validation requirements
- Integrated automated testing and validation workflows to improve coverage, reduce manual execution, and ensure repeatable verification
- Collaborated with cross-functional engineering teams to debug system-level issues, optimize workflows, and deliver production-ready software

PROJECTS

Resume Classification Model

Python, Scikit-learn, Pandas, TF-IDF, Logistic Regression, Flask

- Developed an end-to-end NLP classification pipeline using TF-IDF feature extraction and Logistic Regression to classify resumes across 25 job-role categories
- Preprocessed and transformed 962 unstructured resumes into TF-IDF feature representations for supervised multi-class classification
- Achieved 89.48% test accuracy and 88% macro F1-score across 193 held-out test samples, evaluating model performance using precision, recall, and F1-score
- Deployed the trained ML pipeline through a Flask REST API, enabling real-time resume processing and automated job-category predictions

Local LLM-Powered News Research Platform

Python, LangChain, Streamlit, FAISS, Ollama

- Built an end-to-end NLP data pipeline using Python and WebBaseLoader to extract, clean, chunk, and preprocess unstructured web data for downstream analysis
- Generated high-dimensional text embeddings using nomic-embed-text and implemented FAISS-based vector similarity search for semantic retrieval across news documents
- Developed a retrieval-augmented generation (RAG) pipeline using LangChain and local Llama 3.2 models to perform semantic analysis and generate context-aware responses from processed news data
- Applied text preprocessing, embedding-based feature representation, and similarity-based retrieval to identify relevant information across large collections of unstructured documents